Angles of triangles

1 The measures add to 180°.

2

a $x = 84$

b $w = 65$

c $w = 39$

d $x = 31$

e $x = 33$

3 The sum of the angle measures exceeds 180°.

4 50°

5 Isosceles

6

a No

b No

c Yes

d No

7

a $k = 67$

b $x = 68$

c $k = 51$

d $m = 126$

8

a $x = 64, y = 32$

b $x = 38, y = 32$

c $x = 59, y = 121$

d $x = 60, y = 65$

9

a $x = \frac{149}{9}^\circ$

b $\frac{149}{3}^\circ$

10

a $x = 8$

b 68°

11

a $x = 30$

b 54°

12

a $x = 20$

b 46°

13

a They are corresponding angles.

b They are vertical angles.

c $x = 41$

14

a They are alternate interior angles.

b $x = 24$

Additional questions

15

a $x = 55$

b $x = 40$

c $x = 69$

d $x = 56$

16

a 50°

b 90°

c 27°

d $105^\circ - x^\circ$

17

a $5, 6, 7$

b $\angle 1$

c Exterior angles 5 and 6 are both exterior angles of angle 3. They are also vertical angles, which means they are congruent.

d The measure of an exterior angle of a triangle is equal to the sum of the measures of the two remote interior angles of the triangle. This means angle 5 has the same measure as interior angles 1 and 2.

18

a $x = 67$

b $x = 56$

c $x = 49$

d $x = 44$

19

a $x = 74, y = 43$

b $a = 60, b = 13$

c $a = 61, b = 29$

d $x = 61, m = 138$

20

a

i $x=8$

ii 68°

iii 54°

iv 58°

b

i $x=0.59$

ii 18.36°

iii 2.95°

iv 13.95°

c

i $x=30$

ii 71°

iii 54°

iv 55°

21

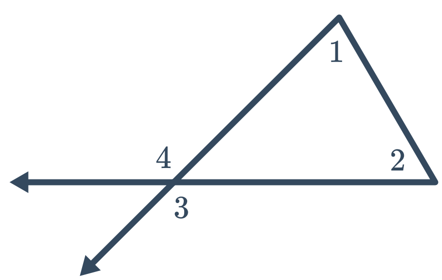
a False, as seen by finding the sum of any non-right triangle, such as an equiangular triangle. The interior angles of any triangle sum to 180° by the triangle sum theorem.

b True, this is the triangle exterior angle theorem.

c False, the triangle sum theorem states that the sum of the measures of the interior angles of a triangle is 180°.

d False, as seen in any right triangle that is not an isosceles right triangle. In a right triangle, the right angle is 90° and the remaining two angles are always complementary. They are only congruent in an isosceles triangle.

22 Yes, this is true. The exterior angles are vertical angles so they are also conguent. In the diagram below, the remote interior angles $\angle 1$ and $\angle 2$ have two exterior angles, $\angle 3$ and $\angle 4$



23 The smallest exterior angle has a measure of 95°. The third internal angle has a measure of $180-37-85=58$ and we know the measure of an exterior angle of a triangle is equal to the sum of the measures of the two remote interior angles of the triangle. The smallest exterior angle will therefore be the sum of the two smallest interior angles which gives us $58+37=95^{\circ}$.